DO NOW — RECALL

Paper 2 targeted reteaching

Name: _____ Class: _____ Date: _____

1. What is O(log n)?

2. What is a recursive base case?

3. What does a trace table show?

4. Trace search for 3 in sorted [1,3,5,7,9].

DO NOW — CHECK / OBJECTIVES

Correct and improve

Check page 1 in GREEN PEN. Write corrections in full.

1. Model answer

Work growing with the number of halvings.

2. Model answer

Condition stopping further calls.

3. Model answer

Intermediate program states.

4. Model answer

mid2→5, high1, mid0→1, low1, mid1→3; return index1.

By the end, you can…

• Correct binary-search bound updates

• Repair recursive termination

• Justify algorithm complexity with evidence

Vocabulary — know these terms

bounds — Limits on a search interval.

base case — Recursive stopping condition.

auxiliary space — Extra storage beyond input.

NEW LEARNING

Read, annotate and explain

Identify

Locate incorrect condition or state update

Model

Model a correct trace table

Apply

Test new data, including missing target

Evaluate

Explain time and memory consequences

Explain one step in your own words, using the correct technical terms.

New Learning · original lesson diagrams

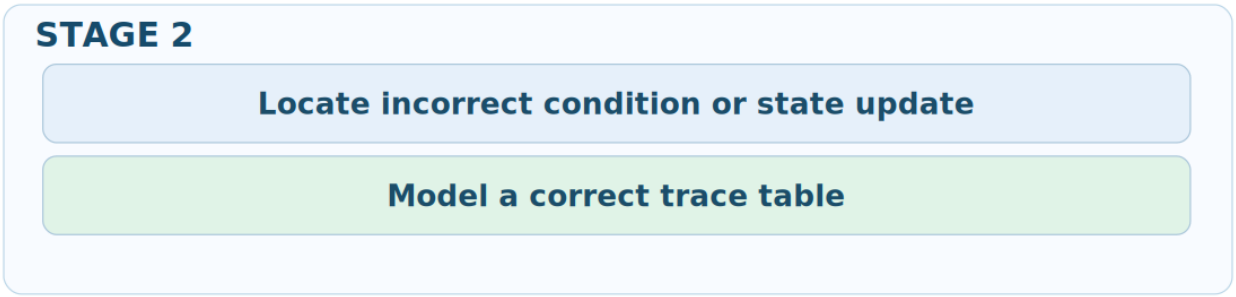
Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



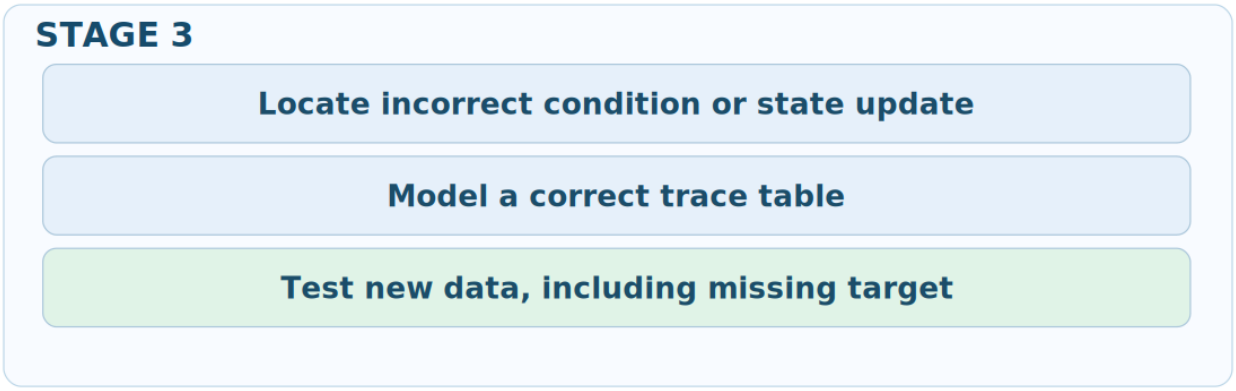
STAGE 1

New Learning



STAGE 2

New Learning

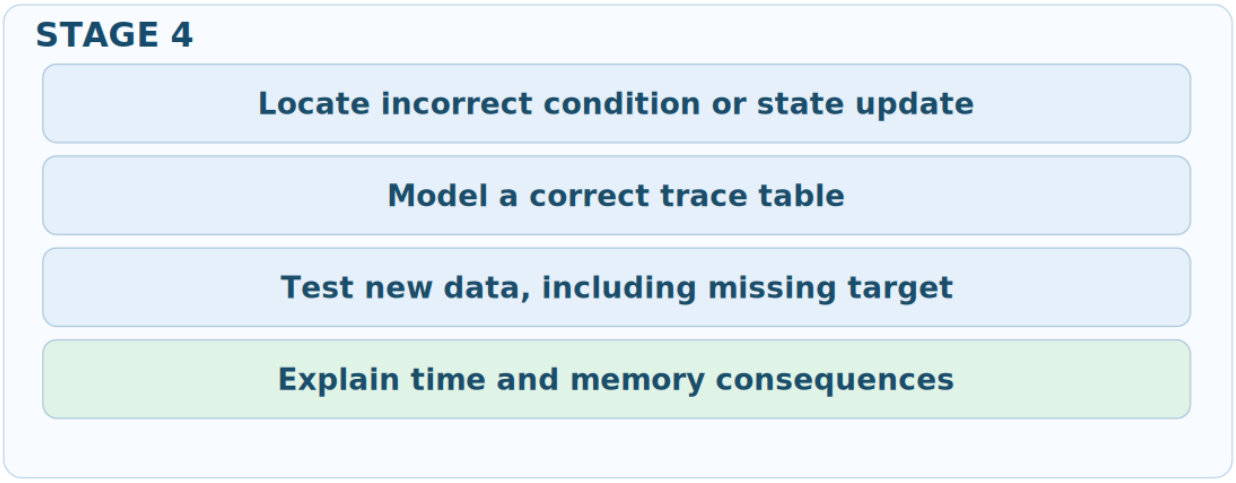


STAGE 3

New Learning · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

New Learning



STAGE 4

I DO — WORKED EXAMPLE

Follow the worked method

Trace search for 3 in [1,3,5,7,9].

Explain the method

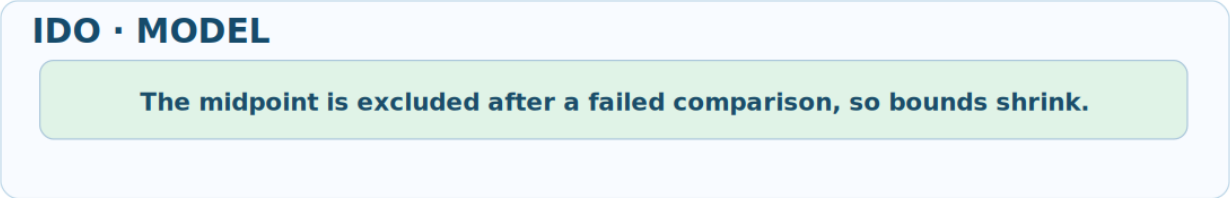
The midpoint is excluded after a failed comparison, so bounds shrink.

I Do · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Trace search for 3 in [1,3,5,7,9].

The midpoint is excluded after a failed comparison, so bounds shrink.



IDO · MODEL

WE DO — GUIDED PRACTICE

Try → Discuss → Improve

TRY: Attempt the guided questions in BLACK PEN. DISCUSS: Compare approaches with a partner, explain your reasoning and agree one improvement. IMPROVE: After the Exit Ticket, use the source CHECK model in GREEN PEN to correct and explain errors.

Guided 1 — TRY

Repair recursive SumTo for n=2.
Source code / input:
if(n==0)return 0; else return n+SumTo(n-1);

YOU DO — INDEPENDENT PRACTICE

Questions · Part 1

Answer independently. Show code, calculations or a justified explanation.

Question 1

Trace binary search for 4 in [2,4,6,8,10], giving bounds/midpoints. [6]

Question 2

Trace recursive SumTo(4), including base0 and return values. [6]

YOU DO — INDEPENDENT PRACTICE

Questions · Part 2

Answer independently. Show code, calculations or a justified explanation.

Question 3

Compare bubble and merge sort worst-case time/extra space; choose for million values with sufficient memory. [6]

EXIT TICKET / REFLECTION

Check your understanding

Complete the actual lesson Exit Ticket before checking.

Binary search may never update the bounds but still always terminate.

Circle: Confident / Getting there / Need help

Explain what you can now do independently.

GREEN PEN — Correct and explain

Choose one error. Write the correct method and explain precisely why your first response was wrong.

WE DO — CHECK AND IMPROVE

Source-grounded model answers

CHECK only after completing the guided practice and Exit Ticket. Use GREEN PEN to improve your method and explain any error.

Guided 1 — CHECK

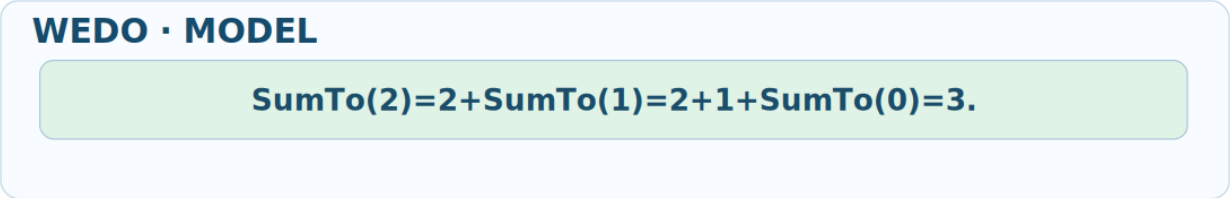
Explain the method
SumTo(2)=2+SumTo(1)=2+1+SumTo(0)=3.

We Do source CHECK · original lesson diagrams

Diagrams appear after the related explanation. Annotate the original source visual.

Repair recursive SumTo for n=2.

SumTo(2)=2+SumTo(1)=2+1+SumTo(0)=3.



WEDO · MODEL